Customer Shopping Behavior Analysis

End-to-End Data Pipeline: Python ETL → MySQL → SQL Analysis → Power BI Dashboard

1. Project Overview

This project builds a complete, end-to-end data analytics pipeline for customer shopping behavior data. It covers the full lifecycle of a real-world analytics workflow: extracting and cleaning raw transactional data with Python, loading it into a MySQL database, performing exploratory data analysis (EDA) and business-transaction queries directly in SQL, and finally connecting Power BI to the database via a native SQL connector to build a live, interactive dashboard.

The dataset consists of 3,900 customer purchase records across 18 columns, capturing customer demographics, purchase details, and shopping behavior patterns.

2. Objective

- Build a reproducible ETL pipeline that takes raw, messy transactional data to a clean, analysis-ready state.
- Centralize the cleaned data in a relational database (MySQL) instead of working off flat files.
- Answer specific business questions using SQL directly on the database.
- Connect Power BI to the live database (not a static export) so the dashboard reflects the source of truth.
- Translate the analysis into concrete, actionable business recommendations.

3. Tech Stack

Layer	Tool / Technology	Purpose
Extraction & Transformation	Python (pandas)	Load raw CSV, clean, impute, engineer features
Database	MySQL	Central store for cleaned data; source of truth for SQL and Power BI
Database Connectivity	mysql-connector-python / SQLAlchemy	Connect Python to MySQL for loading data
Analysis	SQL (MySQL)	EDA and business-transaction queries
Visualization	Power BI Desktop	Interactive dashboard, connected via native MySQL/SQL connector

4. Pipeline Architecture

The pipeline follows a linear, five-stage flow, with MySQL sitting at the center as the single source of truth for both the SQL analysis and the Power BI dashboard:

1. Raw Data (CSV)
3,900 rows × 18 columns of transactional customer data

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2. Python ETL
pandas: clean, impute missing values, standardize columns, engineer features

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3. MySQL Database
Cleaned DataFrame loaded via mysql-connector-python / SQLAlchemy — single source of truth

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4. SQL Analysis Layer
EDA + 10 business-transaction queries run directly in MySQL

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5. Power BI Dashboard
Connected live to MySQL via SQL connector — no manual data export

5. Phase 1 — ETL Using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed the 37 missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake_case for readability and consistency in downstream SQL.
- **Feature Engineering:**
    - Created age_group column by binning customer ages.
    - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified that discount_applied and promo_code_used were redundant; dropped promo_code_used.

6. Phase 2 — Connecting Python to MySQL

Once the DataFrame was clean, it was loaded into MySQL so that all subsequent analysis could run as SQL against a real database rather than in-memory pandas operations. This is also what makes the data available to Power BI later — the database, not the CSV, becomes the shared source of truth.

- Connected to MySQL from Python using SQLAlchemy / mysql-connector-python.
- Created a table matching the cleaned schema (customer_id, age, gender, item_purchased, category, purchase_amount, location, size, color, season, review_rating, subscription_status, shipping_type, discount_applied, previous_purchases, payment_method, frequency_of_purchases, age_group).
- Loaded the cleaned DataFrame into MySQL using df.to_sql(), replacing the need to re-run Python cleaning for every analysis.

7. Phase 3 — EDA and Business Transactions in SQL

With the data in MySQL, we answered ten business questions directly in SQL:

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

Gender	Revenue
Female	75191
Male	157890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount. (839 matching rows)

Customer ID	Purchase Amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

3. Top 5 Products by Rating

Found products with the highest average review ratings.

Item Purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

Shipping Type	Average Purchase Amount
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Subscription Status	Total Customers	Avg Spend	Total Revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3116
New	83
Returning	701

8. Top 3 Products per Category

Listed the most purchased products within each category.

Item Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150

3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Checked whether customers with more than 5 purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2518
Yes	958

10. Revenue by Age Group

Calculated total revenue contribution of each age group.

Age Group	Total Revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

8. Phase 4 — Connecting Power BI to MySQL

Power BI was connected directly to the MySQL database using its native SQL connector, rather than importing a static CSV. This keeps the dashboard synced with the same data used for the SQL analysis.

- Get Data → Database → MySQL database, using the MySQL connector built into Power BI.
- Entered the server host and database name, and authenticated with database credentials.
- Selected Import mode to load the cleaned table(s) into Power BI's data model.
- Verified column data types (numeric fields, dates, categorical fields) after import.
- Built DAX measures for the KPI cards (Total Customers, Total Revenue, Average Rating, Average Purchase Amount).
- Used slicers (Subscription Status, Gender, Category, Shipping Type) so the dashboard filters interactively without touching the underlying SQL.

9. Dashboard

The resulting Power BI dashboard summarizes customer behavior across revenue, subscriptions, product performance, and geography.

10. Key Insights

- Male customers generate more than double the revenue of female customers ($157,890 vs $75,191).
- Non-subscribers account for the large majority of revenue ($170,436 vs $62,645) and spend almost the same per order as subscribers ($59.87 vs $59.49) — subscription status is not currently linked to higher spend.

- Loyal customers (3,116) vastly outnumber New (83) and Returning (701) segments, meaning most of the base has already converted to loyalty — retention matters more than early-funnel acquisition here.
- Express shipping customers spend slightly more on average than Standard shipping customers ($60.48 vs $58.46).
- Revenue is fairly evenly spread across age groups (Young Adult through Senior), so age is not a strong differentiator on its own.
- Certain products (Hat, Sneakers, Coat) rely heavily on discounts (~49-50% of their purchases), which is a margin risk if discounting is reduced.

11. Business Recommendations

- **Review Discount Policy —** Hat, Sneakers, and Coat are structurally discount-dependent; test demand elasticity before cutting discounts on these specific items.
- **Customer Loyalty Programs —** With 3,116 customers already Loyal, focus retention spend on keeping this segment rather than acquisition.
- **Product Positioning —** Feature top-rated items (Gloves, Sandals, Boots) and top sellers per category (Jewelry, Blouse, Sandals, Jacket) in campaigns.
- **Targeted Marketing —** Prioritize male customers and express-shipping users, who show the strongest revenue and spend signals in this data.
- **Re-examine the Subscription Program —** Before investing in promoting subscriptions, address that avg spend is essentially flat between subscribers and non-subscribers; the current program isn't proven to drive higher spend.

12. Conclusion

This project demonstrates a complete analytics pipeline from raw data to a live business dashboard: Python for ETL and data quality, MySQL as the central, queryable source of truth, SQL for structured business analysis, and Power BI connected live via a SQL connector for visualization. Because Power BI reads directly from MySQL rather than a static file, any future update to the underlying data automatically flows through to the dashboard without manual re-export.